Configuração de Servidor DHCP - Projeto SIEM

Objetivo

Configurar um servidor DHCP em ambiente Linux para fornecer endereçamento IP automático em uma rede interna isolada utilizada para testes do SIEM (Wazuh).

Ambiente

Sistema: Ubuntu Server
Interface WAN: enp0s3 (DHCP)
Interface LAN: enp0s8 (192.168.10.1/24)

Configuração de Rede

Definição das interfaces WAN e LAN no servidor.

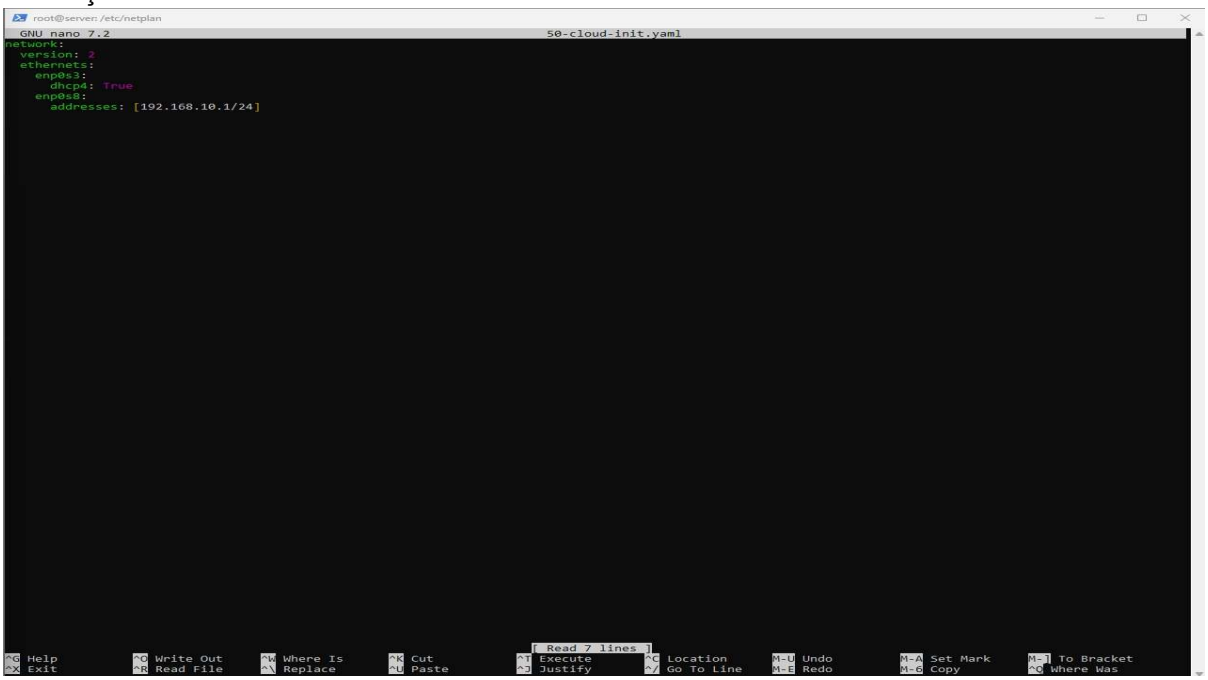


Figura 1 – Configuração do Netplan

/etc/netplan/50-cloud-init.yaml
network: version: 2 renderer: networkd ethernets:
enp0s3:
dhcp4: true enp0s8: addresses:
[192.168.10.1/24]

Instalação DHCP

Instalação do serviço isc-dhcp-server.

```
root@server:/# apt update
Hit:1 https://packages.wazuh.com/4.x/apt stable InRelease
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:3 http://archive.ubuntu.com/ubuntu noble InRelease
Get:4 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,551 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,847 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [250 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [10.7 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:12 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1,166 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.3 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [22.8 kB]
Get:15 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [208 B]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [339 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [177 kB]
Get:18 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.9 kB]
Get:19 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:20 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1,665 kB]
Get:21 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:22 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:23 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7,364 B]
Get:24 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Get:25 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [13.1 kB]
Get:26 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Fetched 7,929 kB in 4s (1,942 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
15 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@server:/#
```

```
root@server:/# apt install isc-dhcp-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  isc-dhcp-common
Suggested packages:
  isc-dhcp-server-ldap policycoreutils
The following NEW packages will be installed:
  isc-dhcp-common isc-dhcp-server
0 upgraded, 2 newly installed, 0 to remove and 15 not upgraded.
Need to get 1,281 kB of archives.
After this operation, 4,281 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 isc-dhcp-server amd64 4.4.3-P1-4ubuntu2 [1,236 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/universe amd64 isc-dhcp-common amd64 4.4.3-P1-4ubuntu2 [45.8 kB]
Fetched 1,281 kB in 3s (459 kB/s)
Preconfiguring packages ...
Selecting previously unselected package isc-dhcp-server.
(Reading database ... 205275 files and directories currently installed.)
Preparing to unpack .../isc-dhcp-server_4.4.3-P1-4ubuntu2_amd64.deb ...
Unpacking isc-dhcp-server (4.4.3-P1-4ubuntu2) ...
Selecting previously unselected package isc-dhcp-common.
Preparing to unpack .../isc-dhcp-common_4.4.3-P1-4ubuntu2_amd64.deb ...
Unpacking isc-dhcp-common (4.4.3-P1-4ubuntu2) ...
Setting up isc-dhcp-server (4.4.3-P1-4ubuntu2) ...
Generating /etc/default/isc-dhcp-server...
Created symlink /etc/systemd/system/multi-user.target.wants/isc-dhcp-server.service → /usr/lib/systemd/system/isc-dhcp-server.service.
Created symlink /etc/systemd/system/multi-user.target.wants/isc-dhcp-server6.service → /usr/lib/systemd/system/isc-dhcp-server6.service.
Setting up isc-dhcp-common (4.4.3-P1-4ubuntu2) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@server:/#
```

Figura 2 – Instalação do pacote DHCP

sudo apt update
sudo apt install isc-dhcp-server -y

Configuração de escopo DHCP

Definição da interface que irá responder às requisições DHCP.

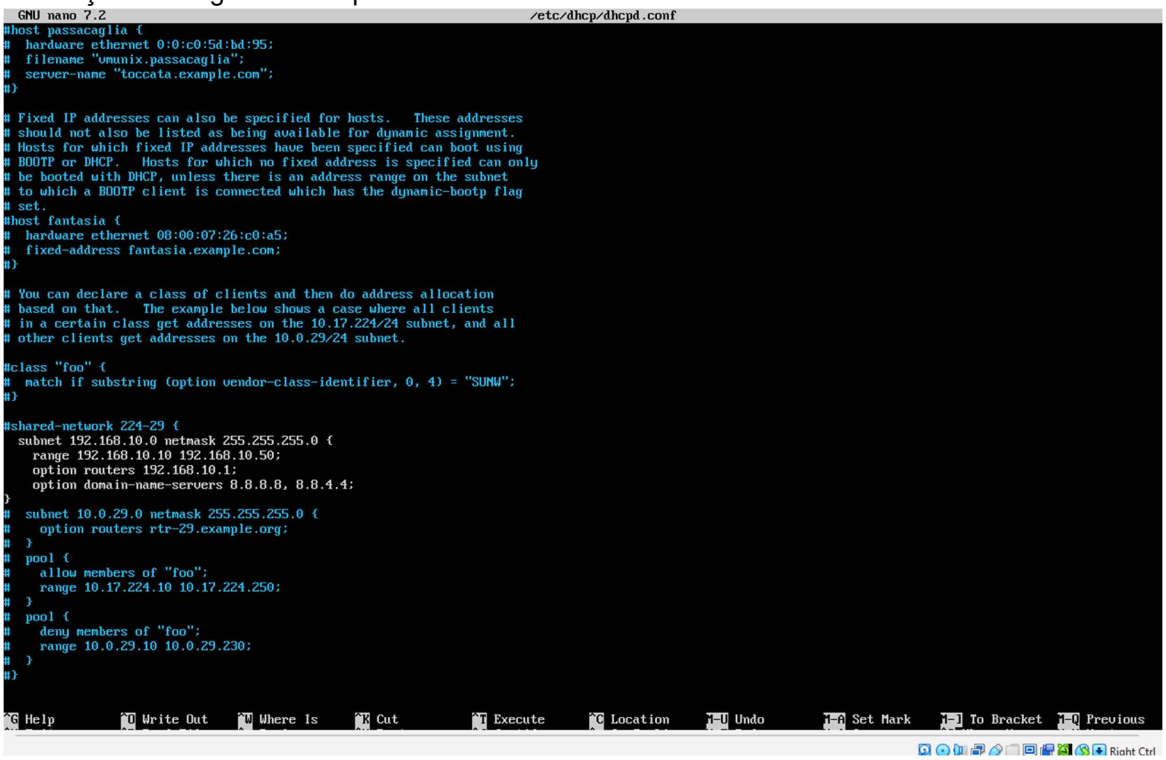
```
  GNU nano 7.2                                  /etc/default/isc-dhcp-server *
# Defaults for isc-dhcp-server (sourced by /etc/init.d/isc-dhcp-server)

# Path to dhcpd's config file (default: /etc/dhcp/dhcpd.conf).
#DHCPDv4_CONF=/etc/dhcp/dhcpd.conf
#DHCPDv6_CONF=/etc/dhcp/dhcpd6.conf

# Path to dhcpd's PID file (default: /var/run/dhcpd.pid).
#DHCPDv4_PID=/var/run/dhcpd.pid
#DHCPDv6_PID=/var/run/dhcpd6.pid

# Additional options to start dhcpd with.
#       Don't use options -cf or -pf here; use DHCPD_CONF/ DHCPD_PID instead
#OPTIONS=""

# On what interfaces should the DHCP server (dhcpd) serve DHCP requests?
#       Separate multiple interfaces with spaces, e.g. "eth0 eth1".
INTERFACESv4="enp0s8"
INTERFACESv6=""
```

Figura 3 – Configuração da interface enp0s8

INTERFACEv4=”enp0s8”

Configuração DHCP

Definição do range de IPs e parâmetros de rede.



```
  GNU nano 7.2                                  /etc/dhcp/dhcpd.conf
#host passacaglia {
#  hardware ethernet 0:0:c0:5d:bd:95;
#  filename "vmunix.passacaglia";
#  server-name "toccata.example.com";
#}

# Fixed IP addresses can also be specified for hosts.   These addresses
# should not also be listed as being available for dynamic assignment.
# Hosts for which fixed IP addresses have been specified can boot using
# BOOTP or DHCP.   Hosts for which no fixed address is specified can only
# be booted with DHCP, unless there is an address range on the subnet
# to which a BOOTP client is connected which has the dynamic-bootp flag
# set.
#host fantasia {
#  hardware ethernet 08:00:07:26:c0:a5;
#  fixed-address fantasia.example.com;
#}

# You can declare a class of clients and then do address allocation
# based on that.   The example below shows a case where all clients
# in a certain class get addresses on the 10.17.224/24 subnet, and all
# other clients get addresses on the 10.0.29/24 subnet.

#class "foo" {
#  match if substring (option vendor-class-identifier, 0, 4) = "SUNW";
#}

#shared-network 224-29 {
  subnet 192.168.10.0 netmask 255.255.255.0 {
    range 192.168.10.10 192.168.10.50;
    option routers 192.168.10.1;
    option domain-name-servers 8.8.8.8, 8.8.4.4;
}
#  subnet 10.0.29.0 netmask 255.255.255.0 {
#    option routers rtr-29.example.org;
#  }
#  pool {
#    allow members of "foo";
#    range 10.17.224.10 10.17.224.250;
#  }
#  pool {
#    deny members of "foo";
#    range 10.0.29.10 10.0.29.230;
#  }
#}
^G Help      ^O Write Out   ^W Where Is   ^K Cut       ^T Execute   ^C Location  M-U Undo      M-A Set Mark  M-] To Bracket M-Q Previous
```

Figura 4 – Configuração do dhcpd.conf

/etc/dhcp/dhcpd.conf

default-lease-time 600; max-lease-time 7200; authoritative;

subnet 192.168.10.0 netmask 255.255.255.0 { range 192.168.10.10 192.168.10.50; option routers 192.168.10.1;

  option domain-name-servers 8.8.8.8, 8.8.4.4; }

Reiniciando serviço e checando status

Service precisa estar como enable e preset também e Active active (running)

```
root@server:/home/server# systemctl restart isc-dhcp-server
root@server:/home/server# systemctl status isc-dhcp-server
● isc-dhcp-server.service - ISC DHCP IPv4 server
     Loaded: loaded (/usr/lib/systemd/system/isc-dhcp-server.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-04-02 00:44:51 UTC; 7s ago
       Docs: man:dhcpd(8)
   Main PID: 4437 (dhcpd)
      Tasks: 1 (limit: 4600)
     Memory: 3.7M (peak: 4.2M)
        CPU: 13ms
     CGroup: /system.slice/isc-dhcp-server.service
             └─4437 dhcpd -user dhcpd -group dhcpd -f -4 -pf /run/dhcp-server/dhcpd.pid -cf /etc/dhcp/dhcpd.conf enp0s8

Apr 02 00:44:51 server dhcpd[4437]: PID file: /run/dhcp-server/dhcpd.pid
Apr 02 00:44:51 server dhcpd[4437]: Wrote 1 leases to leases file.
Apr 02 00:44:51 server sh[4437]: Wrote 1 leases to leases file.
Apr 02 00:44:51 server dhcpd[4437]: Listening on LPF/enp0s8/08:00:27:b1:91:23/192.168.10.0/24
Apr 02 00:44:51 server dhcpd[4437]: Sending on   LPF/enp0s8/08:00:27:b1:91:23/192.168.10.0/24
Apr 02 00:44:51 server sh[4437]: Listening on LPF/enp0s8/08:00:27:b1:91:23/192.168.10.0/24
Apr 02 00:44:51 server sh[4437]: Sending on   LPF/enp0s8/08:00:27:b1:91:23/192.168.10.0/24
Apr 02 00:44:51 server sh[4437]: Sending on   Socket/fallback/fallback-net
Apr 02 00:44:51 server dhcpd[4437]: Sending on   Socket/fallback/fallback-net
Apr 02 00:44:51 server dhcpd[4437]: Server starting service.
root@server:/home/server# _
```

Figura 5 – checagem do serviço DHCP

systemctl restart isc-dhcp-server

systemctl status isc-dhcp-server

Ativação do IP Forwarding

Remover o comentário “#” da linha net.ipv4.ip_forward=1

```
  GNU nano 7.2                                                    /etc/sysctl.conf *
#
# /etc/sysctl.conf - Configuration file for setting system variables
# See /etc/sysctl.d/ for additional system variables.
# See sysctl.conf (5) for information.
#

#kernel.domainname = example.com

# Uncomment the following to stop low-level messages on console
#kernel.printk = 3 4 1 3

###################################################################
# Functions previously found in netbase
#

# Uncomment the next two lines to enable Spoof protection (reverse-path filter)
# Turn on Source Address Verification in all interfaces to
# prevent some spoofing attacks
#net.ipv4.conf.default.rp_filter=1
#net.ipv4.conf.all.rp_filter=1

# Uncomment the next line to enable TCP/IP SYN cookies
# See http://lwn.net/Articles/277146/
# Note: This may impact IPv6 TCP sessions too
#net.ipv4.tcp_syncookies=1

# Uncomment the next line to enable packet forwarding for IPv4
net.ipv4.ip_forward=1

# Uncomment the next line to enable packet forwarding for IPv6
#  Enabling this option disables Stateless Address Autoconfiguration
#  based on Router Advertisements for this host
#net.ipv6.conf.all.forwarding=1


###################################################################
# Additional settings - these settings can improve the network
# security of the host and prevent against some network attacks
# including spoofing attacks and man in the middle attacks through
```

Figura 6 – linha net.ipv4.ip_forward=1 ativa

```
root@server:/# nano /etc/sysctl.conf
root@server:/# root@server:/# sysctl -p
net.ipv4.ip_forward = 1
root@server:/#
```

Figura 7 – aplicar a alteração realizado kernel

sudo sysctl -p

Instalação do iptables-persistent

Instalação necessário pois caso seja preciso reiniciar o servidor as regras de IP table não sejam perdidas.

```
root@server:/# sudo apt install iptables-persistent
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  netfilter-persistent
The following packages will be REMOVED:
  ufw
The following NEW packages will be installed:
  iptables-persistent netfilter-persistent
0 upgraded, 2 newly installed, 1 to remove and 15 not upgraded.
Need to get 14.3 kB of archives.
After this operation, 780 kB disk space will be freed.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 netfilter-persistent all 1.0.20 [7,402 B]
Get:2 http://archive.ubuntu.com/ubuntu noble/universe amd64 iptables-persistent all 1.0.20 [6,946 B]
Fetched 14.3 kB in 1s (11.0 kB/s)
Preconfiguring packages ...
(Reading database ... 205309 files and directories currently installed.)
Removing ufw (0.36.2-6) ...
Skip stopping firewall: ufw (not enabled)
Selecting previously unselected package netfilter-persistent.
(Reading database ... 205214 files and directories currently installed.)
Preparing to unpack .../netfilter-persistent_1.0.20_all.deb ...
Unpacking netfilter-persistent (1.0.20) ...
Selecting previously unselected package iptables-persistent.
Preparing to unpack .../iptables-persistent_1.0.20_all.deb ...
Unpacking iptables-persistent (1.0.20) ...
Setting up netfilter-persistent (1.0.20) ...
Created symlink /etc/systemd/system/iptables.service → /usr/lib/systemd/system/netfilter-persistent.service.
Created symlink /etc/systemd/system/ip6tables.service → /usr/lib/systemd/system/netfilter-persistent.service.
Created symlink /etc/systemd/system/multi-user.target.wants/netfilter-persistent.service → /usr/lib/systemd/system/netfilter-persistent.service.
Setting up iptables-persistent (1.0.20) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@server:/#
```

Figura 7 – instalação iptables-persistent

sudo apt install iptables-persistent

Configuração da rerga de IP Tables para direcionamento do tráfego da rede

Indicado ao firewall para "mascarar" a saída da rede interna para a interface que tem internet (enp0s3).

```
root@server:/home/server# cat /etc/iptables/rules.v4
# Generated by iptables-save v1.8.10 (nf_tables) on Wed Apr  1 00:57:18 2026
*nat
:PREROUTING ACCEPT [0:0]
:INPUT ACCEPT [0:0]
:OUTPUT ACCEPT [0:0]
:POSTROUTING ACCEPT [24:5640]
-A POSTROUTING -o enp0s3 -j MASQUERADE
COMMIT
# Completed on Wed Apr  1 00:57:18 2026
root@server:/home/server#
```

Figura 9 – regra para IP table

sudo iptables -t nat -A POSTROUTING -o enp0s3 -j MASQUERADE

sudo netfilter-persistent save

cat /etc/iptables/rules.v4

Ativação e Testes

```
C:\WINDOWS\system32\cmd.

C:\Users\win11>ipconfig /release

Configuração de IP do Windows


Adaptador Ethernet Ethernet:

   Sufixo DNS específico de conexão. . . . . . :
   Endereço IPv6 de link local . . . . . . . . : fe80::8243:16f1:9d86:fc3a%4
   Gateway Padrão. . . . . . . . . . . . . . . :

C:\Users\win11>ipconfig /renew

Configuração de IP do Windows


Adaptador Ethernet Ethernet:

   Sufixo DNS específico de conexão. . . . . . : example.org
   Endereço IPv6 de link local . . . . . . . . : fe80::8243:16f1:9d86:fc3a%4
   Endereço IPv4. . . . . . . . . . . . . . . : 192.168.10.10
   Máscara de Sub-rede . . . . . . . . . . . . : 255.255.255.0
   Gateway Padrão. . . . . . . . . . . . . . . : 192.168.10.1

C:\Users\win11>

Mostrar ícones ocultos
```

Figura 10 – revodo o IP da máquina cliente

ipconfig /release
ipconfig /renew

```
C:\Users\win11> ping 8.8.8.8

Disparando 8.8.8.8 com 32 bytes de dados:
Resposta de 8.8.8.8: bytes=32 tempo=19ms TTL=117
Resposta de 8.8.8.8: bytes=32 tempo=18ms TTL=117
Resposta de 8.8.8.8: bytes=32 tempo=20ms TTL=117
Resposta de 8.8.8.8: bytes=32 tempo=19ms TTL=117

Estatísticas do Ping para 8.8.8.8:
    Pacotes: Enviados = 4, Recebidos = 4, Perdidos = 0 (0% de
             perda),
Aproximar um número redondo de vezes em milissegundos:
    Mínimo = 18ms, Máximo = 20ms, Média = 19ms

C:\Users\win11>ping google.com

Disparando google.com [172.217.28.142] com 32 bytes de dados:
Resposta de 172.217.28.142: bytes=32 tempo=18ms TTL=118
Resposta de 172.217.28.142: bytes=32 tempo=18ms TTL=118
Resposta de 172.217.28.142: bytes=32 tempo=17ms TTL=118
Resposta de 172.217.28.142: bytes=32 tempo=17ms TTL=118

Estatísticas do Ping para 172.217.28.142:
    Pacotes: Enviados = 4, Recebidos = 4, Perdidos = 0 (0% de
             perda),
Aproximar um número redondo de vezes em milissegundos:
    Mínimo = 17ms, Máximo = 18ms, Média = 17ms

C:\Users\win11>
```

Figura 11 – teste de ping e resolução de DNS

ping 8.8.8.8

ping google.com